CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 5-00-037

WASTE DISCHARGE REQUIREMENTS
FOR
KERN VALLEY HOSPITAL DISTRICT
WASTEWATER TREATMENT FACILITY
KERN COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. Kern Valley Hospital District (hereafter Discharger) owns and operates the Kern Valley Hospital in Mount Mesa, an unincorporated community near Lake Isabella, in Section 26, R33E, T26S, MDB&M, as shown in Attachment A, which is attached hereto and made part of this Order by reference. The facility is within the Kernville hydrologic subarea (No. 554.22), as depicted on hydrologic maps prepared by the Department of Water Resources in August 1986.

2. Wastewater from the hospital is treated and discharged to 20 seepage pits. The treatment plant consists of a rotary screen, flow equalization tank, two stages of trickling filtration, two stages of clarification, and filtration. The treatment plant is designed for a daily flow of 55,125 gallons. Average flow since 1997 has been 17,350 gpd, and the peak flow was 45,000 gpd. Sludge is pumped periodically and disposed of off-site. A diagram of the facilities is shown in Attachment B, which is attached hereto and made part of this Order by reference.

3. The waste is a high-strength wastewater with the following characteristics:

Constituent	Units	Influent	Effluent
BOD_5[1]	mg/l	854 - 120	88.4 - 36
COD[2]	mg/l	770	150
Suspended Solids	mg/l	660	31

[1] Five Day, 20° C Biochemical Oxygen Demand.
[2] Chemical Oxygen Demand.

4. Characterization of wastes from this facility once found a variety of constituents not normally found in domestic wastewater, including several metals. Most of those constituents have since been removed from the waste stream through source-control measures, but a few metals may still appear in the wastewater.

5. The seepage pits are five feet in diameter, approximately 15 to 30 feet in depth, and predominantly underlain by cobbles, gravel and sand. These soils permit rapid percolation of the waste.

6. Groundwater at the site is approximately 60 feet below the ground surface, is of good to excellent quality (EC of 650 µmhos/cm, pH of 7.7) except for the presence of naturally occurring arsenic. Groundwater flows generally to the north towards Lake Isabella.

7. An investigation of groundwater conditions at the site revealed that past operations at the site had adversely impacted groundwater. The main constituent of concern was salt, with EC levels within the impacted area as high as 2,400 µmhos/cm. The principal source was identified as regeneration brine from the ion exchange process used to soften the supply water. The source has since been separated out and the brine concentrate is hauled to the McKittrick Waste Treatment Site for disposal.

8. Average EC of the source water is 815 µmhos/cm. The monthly average EC of the source water varies seasonally from 635 µmhos/cm in July to 940 µmhos/cm in December.

9. Since September 1998, EC of the wastewater has averaged 1,125 µmhos/cm and has exceeded the basin plan limit of source EC plus 500 µmhos/cm on only five days.

10. The *Water Quality Control Plan for the Tulare Lake Basin, Second Edition* (hereafter Basin Plan), designates beneficial uses, establishes water quality objectives, and contains implementation plans and policies for all waters of the Basin. These requirements implement the Basin Plan.

11. The beneficial uses of underlying groundwater are domestic, industrial, and agricultural supply.

12. Surface water drainage is to Lake Isabella. The beneficial uses of Lake Isabella are power generation, water contact recreation, noncontact water recreation, warm and cold fresh water habitat, wildlife habitat, and fresh water replenishment. Lake Isabella drains into the Kern River.

13. The Basin Plan requires that each Report of Waste Discharge for land disposal justify why reclamation is not practiced or proposed. Kern Valley Hospital submitted a report dated 27 December 1995, which reviewed opportunities to reclaim its effluent and concluded that reclamation was not feasible for this facility.

14. The conditional discharge as permitted herein is consistent with the anti-degradation provisions of State Water Resources Control Board Resolution No. 68-16, but is complicated due to concurrent passive cleanup necessitated by past discharges. Some degradation of groundwater immediately beneath the discharge area is appropriate provided no degradation of groundwater occurs beyond a specific area defined by predetermined points of compliance. Degradation of underlying groundwater within these points of compliance is consistent with maximum benefit to the people of the State. The hospital is the major health provider in the area and critical to the well being of area residents, and the hospital has no other means of wastewater disposal. Land use at this location is not expected to change. Assimilative capacity was once available in the underlying groundwater to allow for some degradation that would not unreasonably affect present and anticipated beneficial use of such water or result in water quality less than that described in the Basin Plan. Reasonable degradation if allowed should be limited to only the groundwater within an existing network of monitoring wells located downstream of the disposal area. Presently, the hospital is engaged in a low-level cleanup effort of its past practices under the enforcement oversight of this Board. Groundwater quality exceeds acceptable receiving water quality and must be restored to background quality at the point of compliance and to water quality objectives of the Basin Plan beneath the disposal area. The continuing discharge permitted herein cannot be

allowed to interfere with or otherwise delay that restoration. Groundwater monitoring is required to assure restoration to groundwater quality limitations and subsequent protection of the groundwater to those limitations. Restoration will occur under directives of Cleanup or Abatement Order No. 94- 712, which may be amended hereafter by the Board as appropriate, and subsequent protection will occur under conditions prescribed in this Order.

15. The Facility described herein provides best practicable treatment and control for the subject wastewater, and will assure that the continuing discharge will not interfere with restoration of the aquifer, or create a subsequent condition of pollution or create a condition of nuisance. The Facility, if properly operated and maintained, will assure that the highest water quality will eventually be achieved and maintained. The treatment process incorporates technology for secondary treatment of this wastewater, provides for appropriate biosolids handling, and may require a disinfection process for bacteria and viruses should monitoring establish the need. Control measures include the separate collection and handling of brine waste from water softeners. It also includes a current O&M manual and sufficient staffing and staff training to assure proper source control, operation and maintenance. Conditions of discharge that require this best practicable treatment and control are contained herein.

16. On 19 March 1985, Kern County adopted a mitigated Negative Declaration for this project in accordance with the California Environmental Quality Act (CEQA) (Public Resources Code Section 21000, and following) and the State CEQA Guidelines. The discharge was previously approved by the Board and is exempt from CEQA as an existing facility pursuant to Title 14, California Code of Regulations (CCR), Section 15301.

17. This discharge is exempt from the requirements of Title 27. The exemption, pursuant to Section 20090(b), is based on the following:

    a. The Board is issuing waste discharge requirements,

    b. The discharge, as permitted in this Order, complies with the Basin Plan, and

    c. The wastewater does not need to be managed according to Title 22, California Code of Regulations, Division 4.5, Chapter 11, as a hazardous waste.

18. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.

19. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 94-235 is rescinded and that the Kern Valley Hospital District, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

A. Discharge Prohibitions

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.

2. The surfacing of effluent resulting from discharge to the seepage pits is prohibited.

3. Discharge of waste classified as 'hazardous', as defined in Section 2521(a) of Title 23, CCR, Section 2510, and following, or 'designated', as defined in Section 13173 of the California Water Code, is prohibited.

4. Discharge of brine to the facility is prohibited.

B. Discharge Specifications

1. The daily maximum discharge shall not exceed 55,125 gallons.

2. Effluent concentrations shall not exceed the following limits:

Constituent	Units	Monthly Average	Daily Maximum
BOD_5	mg/l	30	90
Total Suspended Solids	mg/l	30	90
Settleable Solids	ml/l	0.1	0.5

3. If notified pursuant to Provision E.4, the most probable number of total coliform organisms per 100 milliliters in effluent shall not exceed a 7-day median of 23 or a daily maximum of 240.

4. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment and disposal areas.

5. The treatment facilities shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.

6. The maximum EC of the discharge shall not exceed the EC of the source water plus 500 µmhos/cm. The EC of the source water shall be determined as the average EC of the source water for the month in which the effluent samples are taken.

7. Public contact with wastewater shall be precluded through such means as fences and signs or acceptable alternatives.

C. Sludge Disposal Specifications

1. Collected sludges, scums, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Executive Officer and consistent with *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 20005, and following.

2. Any proposed change in approved sludge use or disposal practice shall be reported to the Executive Officer **at least 90 days in advance** of the change.

D. Groundwater Limitations

At the points of compliance (POCs) and pursuant to a schedule set forth in Cleanup or Abatement Order No. 94-712, including any subsequent amendments and revisions, the Discharger shall achieve and maintain the following groundwater criteria:

1. The discharge shall not cause groundwater passing (as measured at) the POCs to contain waste constituents in concentrations statistically greater than background water quality.

2. The discharge, in combination with other sources, natural and otherwise, shall not cause groundwater underlying the disposal area inside the POCs, as measured at internal POCs, to exceed water quality objectives for groundwater as set forth in Chapter III of the Basin Plan.

E. Provisions

1. The Discharger shall comply with Monitoring and Reporting Program No. 5-00-037, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.

2. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements," dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."

3. If, for any reason, the Board determines through the cleanup action that any term of this Order deserves reconsideration due to a change in the facts set forth in the findings, this Order may be reopened to consider the new information.

4. Should groundwater monitoring determine that the population of total coliform organisms in groundwater exceeds water quality objectives, the Discharge shall be notified by the Board and shall, three months from the date of written notification from the Board, have installed and operating a reliable disinfection system sufficient to consistently comply with Effluent Limitation No. B.3.

5. In the event of any change in control or ownership of land or waste discharge facilities described herein, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.

   To assume operation under this Order, the succeeding owner or operator must apply in writing to the Executive Officer requesting transfer of the Order. The request must contain the requesting entity's full legal name, the State of incorporation if a corporation, the name and address and telephone number of the persons responsible for contact with the Board, and a statement. The statement shall comply with the signatory paragraph of Standard Provision B.3

and state that the new owner or operator assumes full responsibility for compliance with this Order. Failure to submit the request shall be considered a discharge without requirements, a violation of the California Water Code. Transfer shall be approved or disapproved by the Executive Officer.

6. The Discharger shall use best practicable treatment and control to comply with this Order.

7. The Discharger must comply with all conditions of this Order at all times, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.

8. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.

9. If reclaimed water is used for construction purposes, it shall comply with the most current edition of "Guidelines for Use of Reclaimed Water for Construction Purposes." Other uses of reclaimed water not specifically authorized herein shall be subject to the approval of the Executive Officer and shall comply with 22 CCR, Division 4.

10. The Board will review this Order periodically and will revise requirements when necessary.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 17 March 2000.

GARY M. CARLTON, Executive Officer

SWG:swg

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 5-00-037
FOR
KERN VALLEY HOSPITAL DISTRICT
WASTEWATER TREATMENT FACILITY
KERN COUNTY

Specific sample station locations shall be established with concurrence of the Board's staff. A description of the stations shall be submitted to Board staff and attached to this Program.

INFLUENT MONITORING

Influent samples shall be collected at the inlet of the headworks at approximately the same time as effluent samples. Influent monitoring shall include at least the following:

Constituents	Units	Type of Sample	Sampling Frequency
Biochemical Oxygen Demand (BOD_5)	mg/l	Grab	Monthly
Total Suspended Solids	mg/l	Grab	Monthly
Barium	µg/l	Grab	Monthly
Zinc	µg/l	Grab	Monthly

EFFLUENT MONITORING

Except for flow, which although included in influent monitoring, may be measured either at the headworks or treatment unit, effluent samples shall be collected just prior to discharge to the disposal facility. Effluent samples shall be representative of the volume and nature of the discharge. Time of collection of a grab sample shall be recorded. Effluent monitoring shall include at least the following:

Constituents	Units	Type of Sample	Sampling Frequency
Flow	mgd	Metered	Continuous
Specific Conductivity	µmhos/cm	Grab	Daily
pH	pH Units	Grab	Daily
Settleable Solids	ml/l	Grab	Daily
Total Dissolved Solids	mg/l	Grab	Monthly
Ammonia	mg/l	Grab	Monthly
Nitrate	mg/l	Grab	Monthly

Constituents	Units	Type of Sample	Sampling Frequency
BOD_5	mg/l	Grab	Weekly
Total Suspended Solids	mg/l	Grab	Weekly
Standard Minerals[1]	mg/l	Grab	Annually
Total Coliform	MPN/100 ml	Grab	Weekly[2]

1 Standard Minerals shall include all major cations and anions and include a verification that the analysis is complete (i.e., a cation/anion balance).

2 Required upon date that disinfection is required, pursuant to notification under Provision E.4 of this Order.

GROUNDWATER MONITORING

Monitoring Network

The "Point of Compliance" for groundwater shall be the vertical surface located at the hydraulic downgradient limit of the disposal area and shall extend through the uppermost aquifer underlying the disposal area. The network shall include a sufficient number of groundwater monitoring wells, installed at appropriate locations and depths to represent the quality of groundwater passing the point of compliance, to monitor restoration of the impacted groundwater, and monitor compliance with limitations thereafter.

The program shall include a sufficient number of background groundwater monitoring wells installed at appropriate locations and depths to represent the quality of groundwater unaffected by discharges from the facility. All well locations and construction features are subject to the review and concurrence of Board staff.

The monitoring well network shall be constructed and maintained as directed by the Board through oversight of Discharger efforts to comply with Cleanup and Abatement Order No. 94-712, including any subsequent amendments and revisions.

Sampling

The monitoring program shall include a detailed description of the sampling and analytical procedures used during monitoring to assure that monitoring results provide a reliable indication of water quality at all background and monitoring points.

The groundwater surface elevation (in feet and hundredths, USGS datum) in all wells shall be measured at the time of sampling (prior to purging) and used to determine the velocity and direction(s) of groundwater flow. This information shall be displayed on a groundwater flow net for the site. All approved monitoring wells shall be sampled and analyzed as follows:

Constituent	Unit	Type of Measurement	Frequency
Electrical Conductivity	µmhos/cm	Field	Quarterly
pH	pH units	Field	Quarterly
Temperature	°C or °F	Field	Quarterly
Total Dissolved Solids	Mg/l	Grab	Quarterly
Nitrate	Mg/l	Grab	Quarterly
Total Coliform	MPN/100 ml	Grab	Quarterly
Total Organic Carbon	Mg/l	Grab	Quarterly
Metals[1]	µg/l	Grab	Annually
Standard Mineral[2]	Mg/l	Grab	Annually

1 "Metals" shall include barium, iron, manganese, and zinc.
2 Standard Minerals shall include all major cations and anions and include a verification that the analysis is complete (i.e., a cation/anion balance).

SLUDGE MONITORING

If sludge is removed from seepage pits and treatment units, but prior to disposal, a composite sample shall be analyzed, on a dry weight basis, for Total Solids (%), Nitrogen (total, NH_4-N, and NO_3-N), Total Phosphorous, Total Potassium, and totals of specific metals (Pb, Zn, Cu, Ni, Cd, Ba, and Ag). Analytical results shall be submitted to Board staff. Analysis of soluble concentrations of these specific metals shall also be included as needed. If final disposal is proposed to go to land, a technical report analyzing application rates and procedures relative to Department of Health Services' *Manual of Good Practices for Landspreading of Sewage Sludge* and EPA's *Process Design Manual for Land Application of Municipal Sludges* and Title 23, California Code of Regulations, Section 2511(f), shall be completed and submitted to the Executive Officer for approval. The report shall be prepared by a California registered civil engineer experienced in wastewater treatment and disposal.

Sampling records shall be retained for a minimum of five years. A log shall be kept of sludge quantities generated and of handling and disposal activities. The frequency of entries is discretionary; however, the log should be complete enough to serve as a basis for part of the annual report.

WATER SUPPLY MONITORING

A sampling station shall be established where a representative sample of the water supply can be obtained. Water supply monitoring shall include at least the following:

Constituents	Units	Sampling Frequency
Electrical Conductivity @ 25°C	µmhos/cm	Monthly

REPORTING

Monthly monitoring reports shall be submitted to the Regional Board by the **first day of the second month** following sample collection. Quarterly monitoring reports shall be submitted with the monthly monitoring results due the month following the close of each calendar quarter. Annual monitoring results shall be submitted by the **first day of February** each year.

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner that illustrates clearly whether the Discharger complies with waste discharge requirements, including calculation of all averages, etc. The Discharger shall also submit the data in a computer format approved by the Executive Officer.

If the Discharger monitors any pollutant at the locations designated herein more frequently than is required by this Order, the results of such monitoring shall be included in the discharge monitoring report.

The Discharger may also be requested to submit an annual report to Board staff with tabular and graphical summaries of the monitoring data obtained during the previous year. Any such request shall be made in writing. The report shall discuss the corrective actions taken and planed to bring the discharge into full compliance with the waste discharge requirements.

By **1 February of each year,** the Discharger shall submit a written report to Board staff containing the following:

1. The names, titles, certificate grade (if any), and general responsibilities of persons operating and maintaining the wastewater treatment plant.

2. The names and telephone numbers of persons to contact regarding the plant for emergency and routine situations.

3. A certified statement of when the flow meter and other monitoring instruments and devices were last calibrated, including identification of who did the calibration (Standard Provision C.4).

4. A statement whether the current operation and maintenance manual, and contingency plan, reflect the wastewater treatment plant as currently constructed and operated, and the dates when these documents were last reviewed for adequacy.

5. The total quantity of sludge disposed of during the previous year and ultimate disposal site(s).

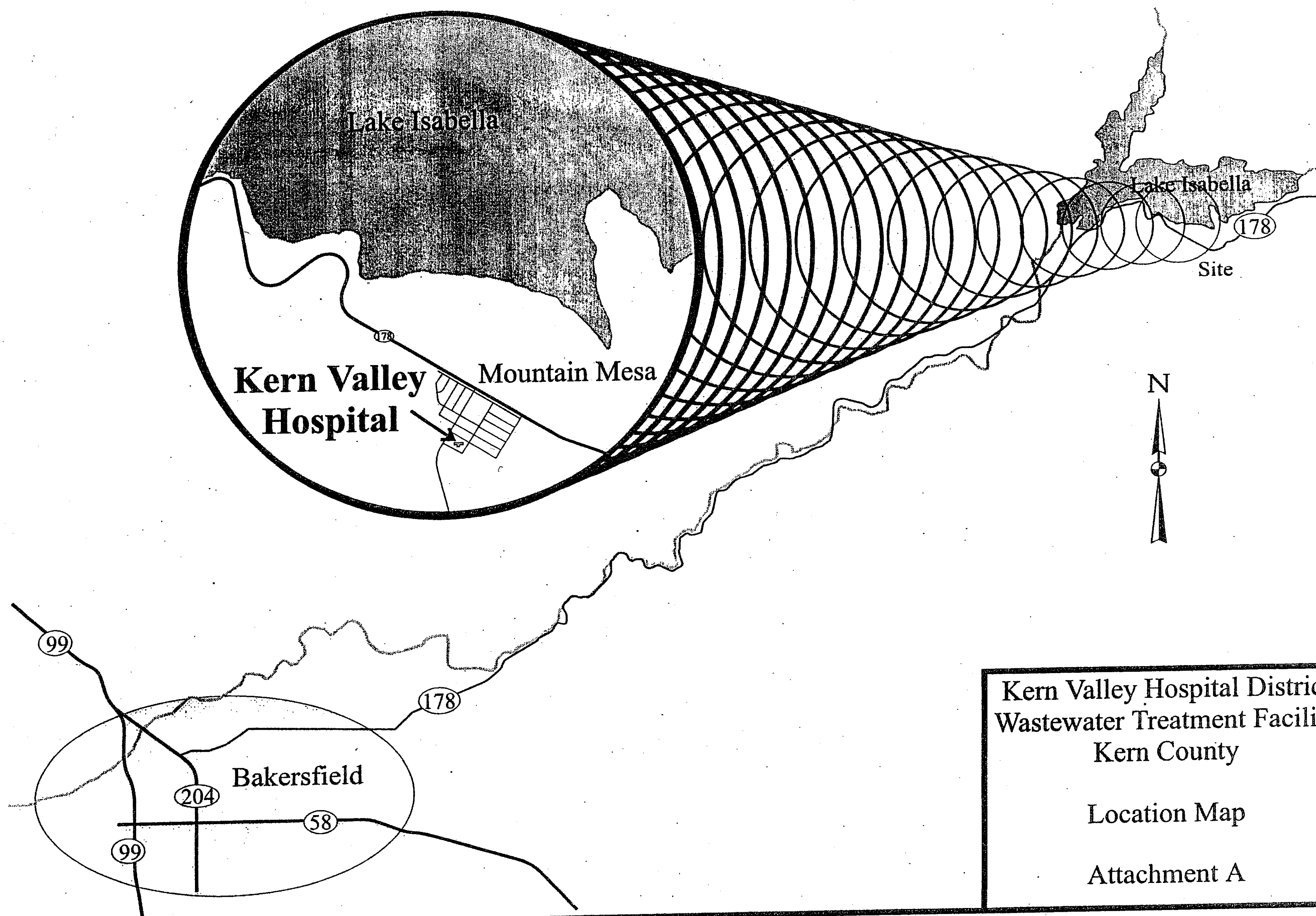
All reports submitted in response to this Order shall comply with the signatory requirements in Standard Provision B.3.

The Discharger shall implement the above monitoring program on the first day of the month following issuance of this program.

Ordered by: ______________________
GARY M. CARLTON, Executive Officer

17 March 2000
(Date)

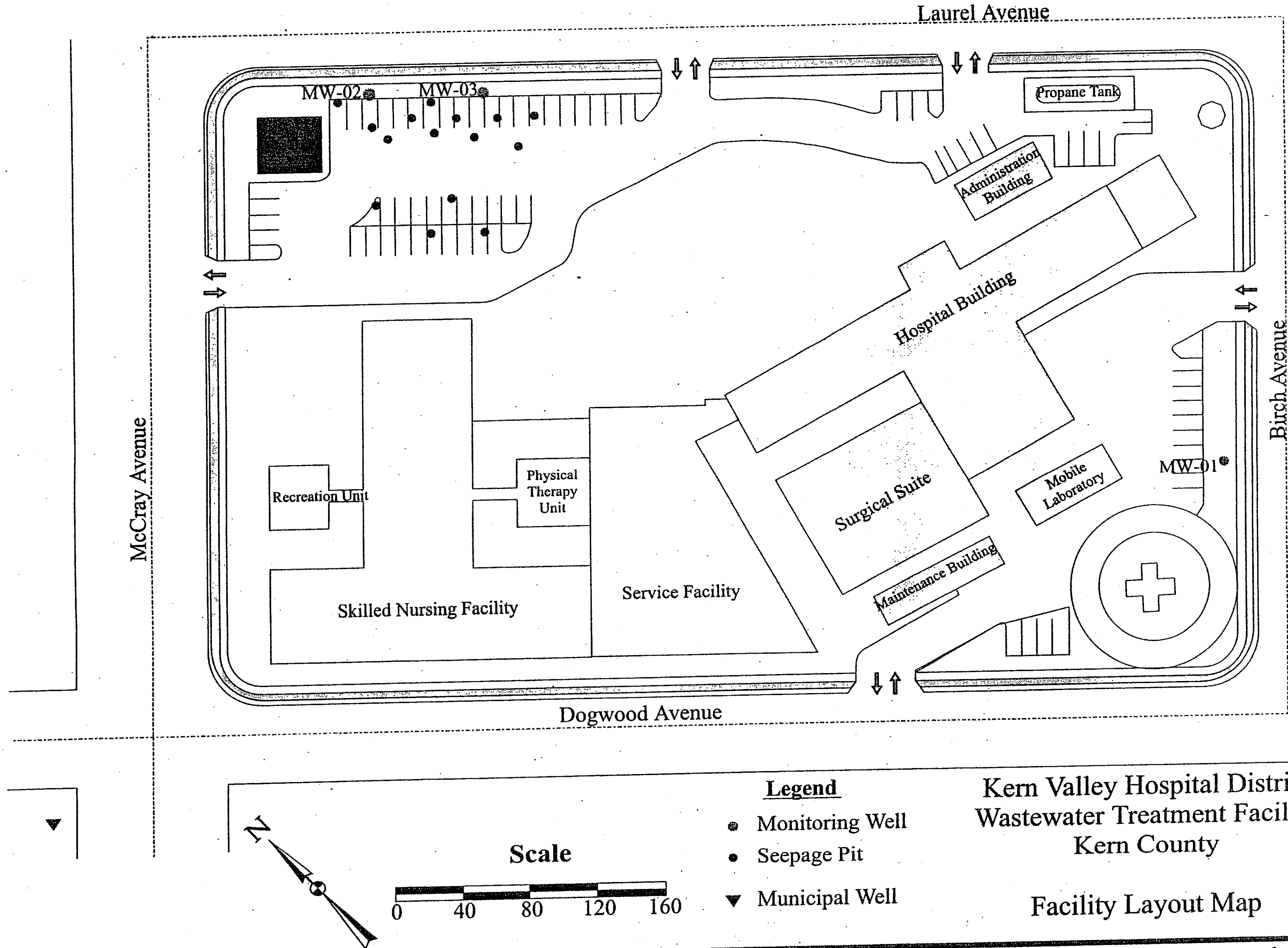
SWG:swg



Kern Valley Hospital District
Wastewater Treatment Facility
Kern County

Location Map

Attachment A



Scale

0 40 80 120 160

Legend

- Monitoring Well
- Seepage Pit
- Municipal Well

Kern Valley Hospital District
Wastewater Treatment Facility
Kern County

Facility Layout Map

Attachment B

INFORMATION SHEET

ORDER NO. 5-00-037
KERN VALLEY HOSPITAL DISTRICT
WASTEWATER TREATMENT FACILITY
KERN COUNTY

Kern Valley Hospital District submitted a Report of Waste Discharge for the discharge of 31,000 gallons per day of wastewater from a wastewater treatment facility to seepage pits.

Characterization of wastes from this facility found a variety of constituents not normally found in domestic wastewater, including several metals. Most of those constituents have since been removed from the waste stream, but a few metals may still appear in the wastewater.

Groundwater has been polluted by discharges from the facility. Affected groundwater has an EC of approximately 2,000 µmhos/cm and a chloride concentration of 225 mg/l.

Groundwater at the site is approximately 60 feet below the ground surface, is of good to excellent quality (EC of 700 µmhos/cm, pH of 7.6) except for the presence of naturally occurring arsenic. Groundwater flows generally to the north towards Lake Isabella. The beneficial uses of Lake Isabella are power generation, water contact recreation, noncontact water recreation, warm and cold fresh water habitat, wildlife habitat, and fresh water replenishment.

An investigation of groundwater conditions at the site revealed that operations at the site had adversely impacted groundwater. The main constituent of concern was salt, with EC levels within the impacted area as high as 2,400 µmhos/cm. The principal source was identified as regeneration brine from the ion exchange process, used to soften the supply water. The source has since been separated out and is no longer discharged to the wastewater treatment plant.

Average EC of the source water is 815 µmhos/cm. The monthly average EC of the source water varies seasonally from 635 µmhos/cm in July to 940 µmhos/cm in December.

Since September 1998, EC of the wastewater has averaged 1,125 µmhos/cm and has exceeded the basin plan limit of source EC plus 500 µmhos/cm twice.

Groundwater monitoring is required by the proposed Order due to the hydrogeologic characteristics of the site (abundant sand and gravel) and the characteristics of the waste.

The County adopted a Negative Declaration for the proposed project in March 1985 pursuant to the California Environmental Quality Act (CEQA) (Public Resources Code Section 21000, et seq.) and the State CEQA Guidelines. The present discharge is exempt from CEQA as an existing facility.

SWG:swg:3/17/2000